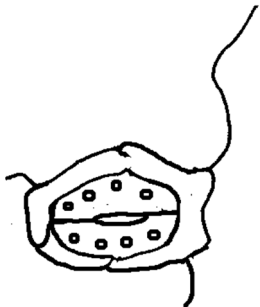


Question Number	Answer	Additional Guidance	Mark
3(a)(i)	A drawing showing the following features: • chloroplasts shown in each guard cell (1) • pore clearly shown (1) • cell walls shown correctly (1)	Example of drawing: 	(3)

Question Number	Answer	Additional Guidance	Mark
3(a)(ii)	An answer showing the following steps: - pore measured (3 epgu), length of pore in metres calculated (1) - conversion to μm done correctly (1) OR - epgu converted to μm correctly (1) $3 \mu\text{m}$ multiplied by number of epgu correctly (1)	ecf applies Example of calculation: $3 \times 10^{-6} \times 3 = 9 \times 10^{-6} \text{ m}$ therefore, pore is $9 \mu\text{m}$ $3 \times 10^{-6} = 3 \mu\text{m}$ $3 \times 3 \times 10^{-6} = 9 \mu\text{m}$	(2)

Question Number	Answer	Additional Guidance	Mark
3(b)(i)	correct scale shown on x axis one 1cm box = 250 ppm (1)		(1)

Question Number	Answer	Additional Guidance	Mark
3(b)(ii)	standard deviations correctly plotted (1)		(1)

Question Number	Answer	Additional Guidance	Mark
3(b)(iii)	An answer that makes reference to the following points: - the SDs for pore area at the two values do not overlap (1) - at 2550 ppm lowest area is 4700, at 1300 highest is 4260 (1)		(2)

Question Number	Answer	Additional Guidance	Mark
3(b)(iv)	An answer showing the following steps: - relevant figures substituted into the equation (1) - correct answer given (1)	<u>Example of calculation:</u> $y = (2.8 \times 6500) - 397.5$ $= 17802.5 \text{ (}\mu\text{m}^2\text{)}$	(2)